

# Cardiovascular and renal outcomes of GLP-1 receptor agonists in type 2 diabetes mellitus: a systematic review and meta-analysis

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## Aims

Glucagon-like peptide-1 receptor agonists (GLP-1RAs) have emerged as a foundational therapeutic option for type 2 diabetes mellitus (T2DM), offering glycaemic control and cardiovascular benefits. This systematic review and meta-analysis evaluate the impact of GLP-1RAs on cardiovascular outcomes in patients with T2DM.

## Methods and results

We conducted a comprehensive search of PubMed, Embase, and Cochrane Library up to 31 March 2025, for randomized controlled trials (RCTs) comparing GLP-1RAs with placebo in T2DM patients. Risk ratios (RR) with 95% confidence intervals (CIs) were pooled using a random-effects model. Twenty RCTs involving 83 004 patients were included. GLP-1RAs significantly reduced the risk of major adverse cardiovascular events (RR 0.87, 95% CI 0.83–0.92), all-cause death (RR 0.89, 95% CI 0.84–0.93), cardiovascular death (RR 0.88, 95% CI 0.81–0.94), myocardial infarction (RR 0.87, 95% CI 0.79–0.96), stroke (RR 0.88, 95% CI 0.81–0.96) and composite renal outcome (RR 0.80, 95% CI 0.73–0.88) compared to placebo. In addition, a trend of reduction in heart failure hospitalizations (RR 0.93, 95% CI 0.85–1.01), and coronary revascularization (RR = 0.87, 95% CI 0.74 to 1.01) was observed with GLP-1 RAs without reaching statistical significance.

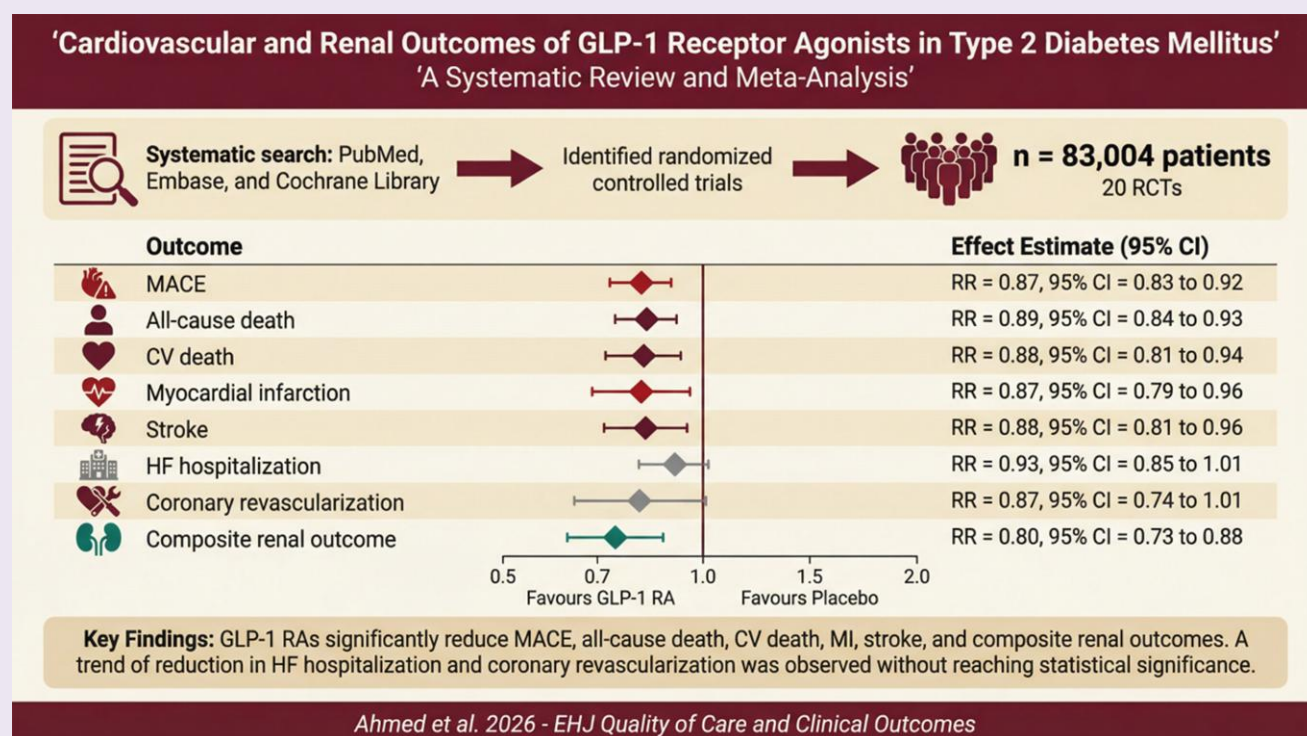
## Conclusion

GLP-1RAs are associated with improved cardiovascular outcomes in patients with T2DM. These findings support current guideline recommendations and highlight their cardioprotective benefits beyond glycaemic control in patients with T2DM.

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## Graphical abstract



## Keywords

GLP-1 receptor agonists • Type 2 diabetes • Cardiovascular outcomes • Meta-analysis

## Key Learning Points

## What is already known

- GLP-1RAs are recommended as second-line therapy after metformin and as preferred first injectable therapy for patients with inadequate glycaemic control, primarily based on their dual benefits of glucose lowering and weight reduction.
- GLP-1 receptor agonists have demonstrated cardiovascular benefits in individual large-scale cardiovascular outcomes trials, but a comprehensive synthesis of all available evidence was lacking.
- Previous meta-analyses examining GLP-1RAs and cardiovascular outcomes were limited by smaller sample sizes and reduced statistical power.

## What this study adds

- This comprehensive meta-analysis of 20 randomized controlled trials involving 83 004 patients demonstrated significant cardiovascular benefits of GLP-1RAs, with high certainty of evidence for major adverse cardiovascular events (MACE), all-cause and cardiovascular mortality.
- Beyond traditional atherosclerotic cardiovascular endpoints, GLP-1RAs provide significant benefits for composite renal outcomes with consistent effects across diverse patient populations.
- These findings support the expansion of GLP-1RA use beyond glycaemic control to encompass broad cardiovascular and renal risk reduction as a foundational therapeutic strategy in patients with type 2 diabetes and established cardiovascular disease or high cardiovascular risk.

## Introduction

Type 2 diabetes mellitus (T2DM) is a chronic metabolic disorder characterized by sustained hyperglycaemia, arising from a combination of peripheral insulin resistance and impaired insulin

secretion.<sup>1,2</sup> This metabolic dysfunction is associated with a wide range of complications, both microvascular and macrovascular, that contribute substantially to global morbidity and mortality.<sup>2,3</sup>

Among these, cardiovascular disease (CVD) remains the leading cause of death in individuals with T2DM, with approximately

60% of the excess mortality observed in this population attributed to cardiovascular causes.<sup>4</sup> The therapeutic landscape of T2DM has evolved significantly, with the introduction of both oral and injectable agents that extend beyond glycaemic control to address comorbid conditions such as obesity and cardiovascular risk.<sup>5,6</sup> While traditional antidiabetic therapies have demonstrated effectiveness in lowering blood glucose levels, some are associated with adverse effects, including weight gain, which may further worsen insulin resistance and cardiovascular risk profiles.<sup>7</sup>

Glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium-glucose cotransporter-2 inhibitors (SGLT2i) have emerged as important therapeutic classes offering cardiometabolic benefits.<sup>8,9</sup> In particular, GLP-1RAs have demonstrated the ability to reduce major adverse cardiovascular events (MACE), including cardiovascular death, non-fatal myocardial infarction, and non-fatal stroke, as evidenced by several large-scale cardiovascular outcomes trials (CVOTs).<sup>10–12</sup> Mechanistically, GLP-1RAs exert their effects through glucose-dependent insulin secretion, suppression of glucagon release, delayed gastric emptying, and appetite regulation—contributing not only to improved metabolic parameters but also to weight loss and cardiovascular benefit.<sup>13</sup>

Given their dual role in improving glycaemic control and promoting weight reduction, GLP-1 RAs are currently recommended as second-line therapy following metformin, and as the preferred first injectable therapy in patients inadequately controlled with oral agents.<sup>14</sup> The publication of recent randomized controlled trials (RCTs) has expanded the available evidence base.<sup>15–18</sup> Although previous meta-analyses have addressed this topic, they were limited by small sample sizes and reduced statistical power.<sup>19,20</sup> Our study aims to strengthen the current evidence base by systematically synthesizing all trials published to date. In addition to conventional relative effect measures, we calculated absolute risk reductions, conducted a range of subgroup analyses, applied the GRADE framework to evaluate the certainty of evidence, and examined a broader spectrum of clinically meaningful outcomes.

## Methods

This meta-analysis was conducted in accordance with the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).<sup>21</sup> The protocol for this meta-analysis has been registered on PROSPERO (CRD420251029582). As this review did not involve any direct patient data collection or investigation by the authors, no ethical approval was not needed.

### Literature search

We conducted a comprehensive search of the Cochrane Library, PubMed, and Embase databases from their inception to 31 March 2025. The search strategy utilized a combination of targeted keywords and Medical Subject Headings (MeSH) terms, including 'GLP-1 agonist,' 'semaglutide,' 'exenatide,' 'albiglutide,' 'dulaglutide,' 'liraglutide,' 'type 2 diabetes' and 'RCT.' Boolean operators (AND, OR) were applied to optimize the search and ensure the inclusion of all relevant studies. Further details regarding the search terms and strategy are provided in [Supplementary material online, Table S1](#).

### Eligibility criteria

The inclusion criteria for this review were as follows: (i) study design: randomized controlled trials (RCTs); (ii) population: patients aged 18 years and above with type 2 diabetes, (iii) intervention: GLP-1 RA; (iv) comparator: placebo; (v) outcome: reporting at least one relevant clinical outcome of interest.

No restriction was imposed on the type of GLP-1 RA. Studies were excluded if they were not randomized controlled trials (RCTs), including case reports, case series, observational studies, or animal-based studies.

### Study selection, data extraction and outcomes

Rayyan software was utilized to screen and remove duplicates from the studies identified in the online search. Following duplicates removal, two authors (HA and RZ) independently carried out title and abstract screening. Afterwards, a comprehensive full-text review was performed by the same authors. In instances of disagreement, a third author (MA) facilitated resolution.

Study characteristics, such as authors and study locations, inclusion criteria, total population, drug and dosage, sample size, age, sex, BMI, comorbidities (diabetes, hypertension, smoking etc.) were extracted. The primary outcomes included major adverse cardiovascular events (MACE) and all-cause death. The secondary outcomes included cardiovascular death, hospitalization due to HF (HHF), MI, stroke, revascularization, hospitalization for unstable angina, and composite renal endpoint. The criteria used for reporting MACE and the composite renal endpoint are reported in [Supplementary material online, Tables S2 and S3](#).

### Risk of bias and certainty of evidence assessment

The risk of bias for all included trials was assessed using the Cochrane Risk of Bias 2.0 (RoB 2.0) tool.<sup>22</sup> Each included study was assessed for bias and categorized as having either low, high, or some concerns. Two independent reviewers (RZ and TMH) evaluated the risk of bias in each study. Any discrepancies were addressed through discussions with a third reviewer (MA). The certainty of the evidence was evaluated using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) framework. Based on the GRADE Working Group guidelines, the certainty of the pooled estimates was rated as high, moderate, low, or very low.

### Statistical analysis

Statistical analysis was conducted using R version 4.4.1, using the 'meta' package. Risk ratios (RR) with 95% confidence intervals (CIs) were calculated as summary statistics, utilizing a random-effects model and used as primary analysis for inference. The  $I^2$  statistic was used to assess statistical heterogeneity across the trials, with a value of 50% considered indicative of substantial heterogeneity. Absolute risk reduction was estimated by applying pooled RRs to the baseline risk from the control groups. Summary of findings tables were generated using GRADEpro (<https://gdt.gradepro.org/app/>). Publication bias was assessed using a funnel plot for outcomes reported by  $\geq 10$  studies. Egger's test was applied to evaluate funnel plot asymmetry. Leave-one-out sensitivity analyses were performed, where each study was sequentially removed to determine its influence on aggregate findings. Additionally, to account for variation in follow-up durations among the included trials, a sensitivity analysis was conducted in which only studies reporting hazard ratios (HRs) for time-to-event outcomes were retained. Summary HRs and their corresponding 95% confidence intervals (CIs) were estimated using a random-effects model. Subgroup analyses for the primary outcomes were performed according to baseline chronic kidney disease status, trial size ( $<1000$  vs.  $\geq 1000$  participants), type of GLP-1 receptor agonist, route of administration (oral vs. subcutaneous), trial design (cardiovascular outcome trial vs. non-cardiovascular outcome trial), publication era ( $\geq 2018$  vs.  $\leq 2017$ ), and concomitant SGLT2 inhibitor use (enriched [ $\geq 15\%$ ] vs. non-enriched [ $<15\%$ ]). For the composite kidney outcome, subgroup analysis was additionally conducted based on alignment of outcome definitions with Kidney Disease: Improving Global Outcomes (KDIGO) recommendations. Heterogeneity of treatment effects across subgroups was evaluated using interaction tests, which assess whether relative treatment effects differ between strata rather than directly comparing effect sizes. Furthermore, a cumulative meta-analysis was conducted in which studies were

sequentially added in chronological order based on publication year, and pooled effect estimates were recalculated at each step to assess temporal changes in effect estimates. A  $P$ -value  $< 0.05$  indicated statistical significance for all endpoints.

## Results

### Study selection

A total of 3635 records were identified from the databases, including PubMed (431), Cochrane (1671), and Embase (1533). After excluding 1176 duplicate entries, 2459 records were screened based on their titles and abstracts. Of these, 2198 records were excluded and full-texts of 261 articles were reviewed, after which 20 RCTs were deemed eligible for inclusion (see [Supplementary material online, Figure S1](#)).

### Characteristics of included studies

A total of 20 RCTs were included in this analysis.<sup>15–18,23–38</sup> The included studies reported data for 83 004 patients (GLP-1 RA: 42 977 and placebo: 40 027). The weighted mean age of participants was  $63.8 \pm 2.7$  years, with female participants making up 36.0% of the study population. The mean baseline HbA1c levels were  $8.12 \pm 0.5\%$ , and body mass index (BMI) was  $32.3 \pm 1.1$  kg/m<sup>2</sup>. The weighted mean follow-up duration was  $35.6 \pm 16$  months, representing 246 387 total person-years of follow-up. Seven different GLP-1RA agents were evaluated, with semaglutide (subcutaneous and oral formulations) being the most frequently studied (8 studies, 23 011 participants). Further details regarding study design, interventions, and baseline characteristics are summarized in [Table 1](#) and [Supplementary material online, Tables S4 and S5](#). Fourteen trials had a low risk of bias, and 6 had some concerns related to randomization and deviations from intended intervention. The risk of bias assessment is presented in [Supplementary material online, Figure S2](#).

### Primary outcomes

#### MACE

Pooled analysis from 16 RCTs demonstrated that patients receiving GLP-1 RA therapy had a significantly lower risk for MACE compared with placebo (RR = 0.87, 95% CI = 0.83 to 0.92;  $P < 0.0001$ ,  $I^2 = 22.7\%$ , [Figure 1A](#)) [absolute risk difference 20 (95% CI 26 to 12) fewer per 1000 patients; high certainty, [Table 2](#)]. In a sensitivity analysis restricted to 11 trials reporting HRs for time-to-event data, the effect of GLP-1 RA therapy on MACE remained consistent (HR = 0.87, 95% CI = 0.81 to 0.93;  $P < 0.0001$ ;  $I^2 = 43.7\%$ , [Supplementary material online, Figure S3](#)). Upon leave-one-out sensitivity analysis, the overall effect size remained significant ( $P < 0.05$ ), and the  $I^2$  reduced to 5% upon excluding the FREEDOM CVO trial (see [Supplementary material online, Figure S4](#)). Visual inspection of the funnel plot did not suggest asymmetry. Egger's regression test did not indicate evidence of small-study effects ( $P = 0.20$ , [Supplementary material online, Figure S5](#)). No significant subgroup interactions were observed for any of the covariates assessed (all  $P$  for interaction  $> 0.05$ , [Supplementary material online, Figures S6–S12](#)).

#### All-cause death

A pooled analysis of 20 RCTs demonstrated a statistically significant reduction in all-cause death with GLP-1 RAs as compared to placebo (RR = 0.89, 95% CI = 0.84–0.93;  $P < 0.0001$ ,  $I^2 = 0\%$ , [Figure 1B](#)) [absolute risk difference 11 (95% CI 16–7) fewer per 1000 patients; high certainty]. The results remained

consistent on pooling 12 RCTs reporting HRs for all-cause death (HR = 0.88, 95% CI = 0.82–0.94;  $P = 0.0002$ ;  $I^2 = 13.2\%$ , [Supplementary material online, Figure S13](#)). Leave-one-out sensitivity analysis did not reveal any significant findings (see [Supplementary material online, Figure S14](#)). No evidence of publication bias was observed (Egger's test,  $P = 0.11$ , [Supplementary material online, Figure S15](#)). No significant subgroup interactions were observed for any of the covariates assessed (all  $P$  for interaction  $> 0.05$ , [Supplementary material online, Figures S16–S22](#)).

### Secondary outcomes

#### Cardiovascular death

The pooled analysis from 13 RCTs demonstrated significantly reduced risk of CV death in GLP-1 RA group compared to placebo (RR = 0.88, 95% CI = 0.81–0.94;  $P < 0.0006$ ,  $I^2 = 14.4\%$ , [Figure 2A](#)) [absolute risk difference 7 (95% CI 12–4) fewer per 1000 patients; high certainty]. The results remained comparable on pooling 12 RCTs reporting HRs for CV death (HR = 0.87, 95% CI = 0.79–0.94;  $P = 0.0009$ ;  $I^2 = 23.9\%$ , [Supplementary material online, Figure S23](#)). Leave-one-out sensitivity analysis by excluding the PIONEER-6 and FLOW trials reduced the  $I^2$  to 2% and 6%, respectively. (see [Supplementary material online, Figure S24](#)). No asymmetry was noted in the funnel plot (Egger's test,  $P = 0.21$ , [Supplementary material online, Figure S25](#)).

#### Myocardial infarction

The pooled analysis from 13 RCTs demonstrated a statistically significant reduction in the risk of MI with GLP-1 receptor agonists compared to placebo (RR = 0.87, 95% CI = 0.79–0.96;  $P < 0.004$ ,  $I^2 = 45.3\%$ , [Figure 2B](#)) [absolute risk difference 9 (95% CI 15–3) fewer per 1000 patients; high certainty]. The results remained comparable on pooling 11 RCTs reporting HRs for MI (HR = 0.88, 95% CI = 0.80–0.97;  $P = 0.01$ ;  $I^2 = 47.6\%$ , [Supplementary material online, Figure S26](#)). Leave-one-out sensitivity analysis did not reveal any significant findings (see [Supplementary material online, Figure S27](#)). No asymmetry was noted in the funnel plot (Egger's test,  $P = 0.32$ , [Supplementary material online, Figure S28](#)).

#### Stroke

The pooled analysis from 13 RCTs demonstrated a statistically significant reduction in the risk of stroke with GLP-1 receptor agonists compared to placebo (RR = 0.88, 95% CI = 0.81–0.96;  $P < 0.002$ ,  $I^2 = 0\%$ , [Figure 3A](#)) [absolute risk difference 5 (95% CI 7–2) fewer per 1000 patients; high certainty]. The results remained comparable on pooling 11 RCTs reporting HRs for stroke reduction with GLP-1 RA therapy (HR = 0.87, 95% CI = 0.80–0.95;  $P = 0.002$ ;  $I^2 = 5.1\%$ , [Supplementary material online, Figure S29](#)). Leave-one-out sensitivity analysis did not reveal any significant findings (see [Supplementary material online, Figure S30](#)). No asymmetry was noted in the funnel plot (Egger's test,  $P = 0.97$ , [Supplementary material online, Figure S31](#)).

#### Heart failure hospitalization

The pooled analysis from 10 RCTs demonstrated a trend of reduction in the risk of HHF with GLP-1 receptor agonists compared to placebo without reaching statistical significance (RR = 0.93, 95% CI = 0.85–1.01;  $P = 0.09$ ,  $I^2 = 0\%$ , [Figure 3B](#)) [absolute risk difference 4 (95% CI 8 fewer to 1 more) per 1000 patients; moderate certainty]. Similar results were observed on pooling seven RCTs that reported HRs (HR = 0.93, 95% CI = 0.85–1.02;  $P = 0.12$ ;  $I^2 = 0\%$ , [Supplementary material online, Figure S32](#)). Leave-one-out sensitivity analysis did not reveal any significant findings (see [Supplementary material online,](#)





Table 1 Continued

| Study                      | Sample size (N) |         | Age, years <sup>a</sup> |             | Female, % |         | BMI, kg/m <sup>2a</sup> |            | Glycated hemoglobin HbA1c, % <sup>a</sup> |             | Follow-up (months) | GLP-1RA regimen                                         |
|----------------------------|-----------------|---------|-------------------------|-------------|-----------|---------|-------------------------|------------|-------------------------------------------|-------------|--------------------|---------------------------------------------------------|
|                            | GLP-1RA         | Placebo | GLP-1RA                 | Placebo     | GLP-1RA   | Placebo | GLP-1RA                 | Placebo    | GLP-1RA                                   | Placebo     |                    |                                                         |
| Reusch 2014<br>(HARMONY 1) | 150             | 151     | 55.2 ± 9.98             | 54.9 ± 9.40 | 38.7      | 41.7    | 33.6 ± 5.9              | 34.7 ± 5.6 | 8.1 ± 1.0                                 | 8.1 ± 0.9   | 36                 | 30 mg subcutaneous injection of albiglutide once weekly |
|                            |                 |         |                         |             |           |         |                         |            |                                           |             |                    |                                                         |
| Home 2014<br>(HARMONY 5)   | 271             | 115     | 54.5 ± 9.5              | 55.7 ± 9.6  | 50.2      | 39.1    | 32.4 ± 5.5              | 31.8 ± 4.9 | 8.19 ± 0.91                               | 8.26 ± 0.98 | 36                 | Subcutaneous injection of albiglutide 30 mg/week        |
|                            |                 |         |                         |             |           |         |                         |            |                                           |             |                    |                                                         |
| Ahren 2014<br>(HARMONY 3)  | 302             | 101     | 54.3 ± 10.1             | 56.1 ± 10.0 | 55.3      | 50.5    | 32.6 ± 5.5              | 32.0 ± 5.6 | 8.1 ± 0.8                                 | 8.2 ± 0.9   | 24                 | Once-weekly 30→50 mg subcutaneous albiglutide           |
|                            |                 |         |                         |             |           |         |                         |            |                                           |             |                    |                                                         |

BMI, body mass index; CVO, cardiovascular outcomes; DM, diabetes mellitus; GLP-1RA, glucagon-like peptide-1 receptor agonist; HbA1c, glycated haemoglobin; ITCA 650, implantable osmotic mini-pump delivering exenatide; N, number of participants.  
<sup>a</sup>Data are reported as median (interquartile range) or mean ± standard deviation.

Figure S33). No asymmetry was noted in the funnel plot (Egger's test,  $P = 0.65$ , [Supplementary material online, Figure S34](#)).

Coronary revascularization

The pooled analysis from five RCTs demonstrated a trend of reduction in the risk of coronary revascularization with GLP-1 receptor agonists compared to placebo without reaching statistical significance ( $RR = 0.87$ , 95%  $CI = 0.74-1.01$ ;  $P = 0.05$ ,  $I^2 = 67.7\%$ , [Figure 4A](#)) [absolute risk difference 16 (95%  $CI$  33 fewer to 1 more) per 1000 patients; moderate certainty]. However, in sensitivity analysis restricted to four trials reporting hazard ratios, GLP-1 RA therapy was associated with a significant reduction in coronary revascularization ( $HR = 0.81$ , 95%  $CI: 0.69-0.95$ ;  $P = 0.008$ ;  $I^2 = 55.2\%$ , [Supplementary material online, Figure S35](#)). Leave-one-out sensitivity analysis indicated that omission of EXSCEL 2017 reduced heterogeneity to 56.7% and the pooled effect size became statistically significant (see [Supplementary material online, Figure S36](#)).

Hospitalizations due to unstable angina

The pooled analysis from seven RCTs demonstrated no significant difference between GLP-1 RA and placebo for reduction in hospitalizations due to unstable angina ( $RR = 0.99$ , 95%  $CI = 0.86-1.16$ ;  $P = 0.94$ ,  $I^2 = 0\%$ , [Figure 4B](#)) [absolute risk difference 0 (95%  $CI$  4 fewer to 4 more) per 1000 patients; moderate certainty]. The results remained comparable on pooling HRs ( $HR = 0.99$ , 95%  $CI = 0.85-1.15$ ;  $P = 0.87$ ;  $I^2 = 0\%$ , [Supplementary material online, Figure S37](#)). Leave-one-out sensitivity analysis did not reveal any significant findings (see [Supplementary material online, Figure S38](#)).

Composite renal outcome

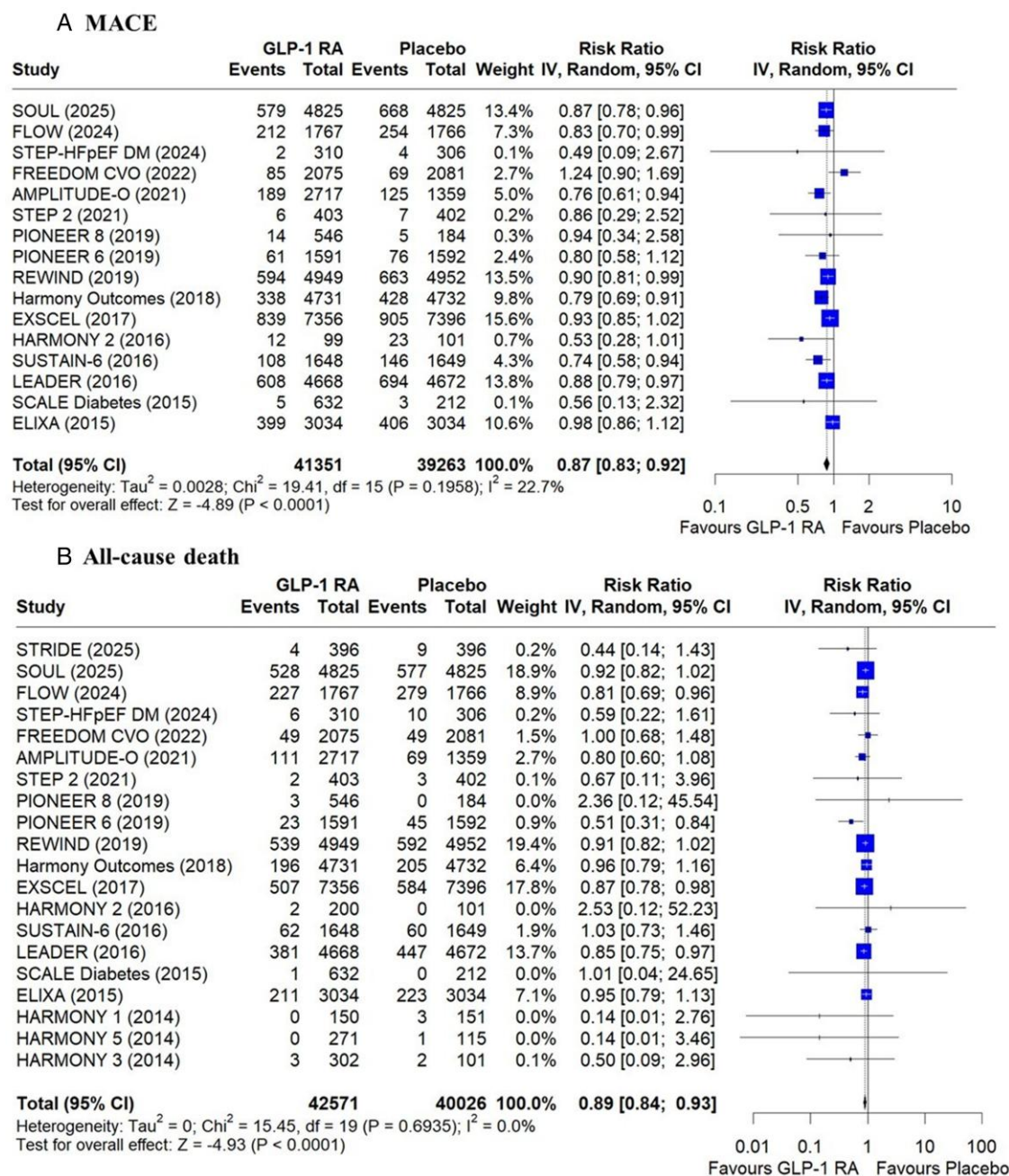
The pooled analysis from eight trials demonstrated a statistically significant reduction in the risk of composite renal outcomes with GLP-1 RAs compared to placebo ( $RR = 0.80$ , 95%  $CI = 0.73-0.88$ ;  $P < 0.0001$ ,  $I^2 = 41.7\%$ , [Figure 4C](#)) [absolute risk difference 23 (95%  $CI$  31 to 14) fewer per 1000 patients; high certainty]. The results remained consistent on pooling 6 RCTs reporting HRs ( $HR = 0.79$ , 95%  $CI = 0.72-0.87$ ;  $P < 0.0001$ ;  $I^2 = 52.6\%$ , [Supplementary material online, Figure S39](#)). Leave-one-out sensitivity analysis did not reveal any significant findings (see [Supplementary material online, Figure S40](#)). No significant subgroup interaction was observed based on alignment with KDIGO criteria for reporting of composite outcome ( $P$  for interaction  $>0.05$ , [Supplementary material online, Figure S41](#)).

The cumulative analysis showed early variability followed by stable and statistically significant risk reduction in MACE and all-cause death after 2016, with consistent direction of effect and increasing precision over time (see [Supplementary material online, Figures S42 and S43](#)).

Discussion

Our meta-analysis comprising a pooled sample of 83 004 patients demonstrated with high certainty that GLP-1 RAs reduced all-cause death, MACE, CV death, stroke, MI, and composite renal outcomes among patients with T2DM. Furthermore, our findings indicated a trend towards a reduction in HHF and coronary revascularization, although these did not reach statistical significance in the primary pooled analysis and the certainty of evidence was rated as moderate.

We observed a significant reduction in all-cause and CV mortality among T2DM patients treated with GLP-1RAs.



**Figure 1** Forest plots of (A) MACE and (B) all-cause death.

Mechanistically, GLP-1RAs enhance glucose-dependent insulin secretion and suppress glucagon, improving glycaemic control and reducing hyperglycaemia-related complications.<sup>39</sup> They also promote satiety and weight loss, which is important given the established link between obesity and cardiovascular disease.<sup>40</sup> Beyond glycaemic effects, GLP-1RAs improve cardiovascular risk factors such as systolic blood pressure and lipid profiles. Feng *et al.*'s network meta-analysis found significant LDL-C reductions across all GLP-1RAs compared with placebo and other antihyperglycemics.<sup>41</sup> Systolic blood pressure reductions range from 1 mmHg with lixisenatide to 5.4 mmHg with semaglutide, possibly mediated via natriuretic and vasodilatory

pathways.<sup>40,42</sup> Anti-inflammatory effects, including suppression of pro-inflammatory cytokines and oxidative stress, also contribute to CV protection.<sup>43</sup>

Second, our analysis demonstrated a significant reduction in MACE, stroke, and MI with GLP-1 RA therapy. These findings align with those of the individual landmark CVOTs. The LEADER trial demonstrated a significant reduction in MACE and all-cause mortality with liraglutide, including a 13% reduction in nonfatal MI and an 11% reduction in nonfatal stroke.<sup>33</sup> AMPLITUDE-O showed a 27% reduction in MACE and a 32% reduction in renal outcomes with efpeglenatide in patients with CVD or CKD.<sup>24</sup> In the REWIND trial, dulaglutide

**Table 2** Summary of findings for GLP-1 RA vs. placebo for T2DM patients

| Outcome                                | Patients (studies), N | Relative effect (95% CI)    | Absolute effects (95% CI) |                                    |                                                            | Certainty       |
|----------------------------------------|-----------------------|-----------------------------|---------------------------|------------------------------------|------------------------------------------------------------|-----------------|
|                                        |                       |                             | Placebo                   | GLP-1 RA                           | Difference                                                 |                 |
| MACE                                   | 80 614<br>(16 RCTs)   | RR = 0.87<br>(0.83–0.92)    | 152 per 1000              | 132 per 1000<br>(126–140)          | 20 fewer per 1000<br>(from 26 fewer to 12 fewer)           | ⊕⊕⊕⊕<br>High    |
| All-cause death                        | 82 597<br>(20 RCTs)   | RR = 0.89<br>(0.84 to 0.93) | 102 per 1000              | 91 per 1000<br>(86 to 95)          | Eleven fewer per 1000<br>(from 16 fewer to 7 fewer)        | ⊕⊕⊕⊕<br>High    |
| CV death                               | 78 457<br>(13 RCTs)   | RR = 0.88<br>(0.81 to 0.94) | 64 per 1000               | 57 per 1000<br>(52 to 60)          | Seven fewer per 1000<br>(from 12 fewer to 4 fewer)         | ⊕⊕⊕⊕<br>High    |
| Myocardial infarction                  | 78 993<br>(13 RCTs)   | RR = 0.87<br>(0.79 to 0.96) | 71 per 1000               | 62 per 1000<br>(56 to 69)          | 9 fewer per 1000<br>(from 15 fewer to 3 fewer)             | ⊕⊕⊕⊕<br>High    |
| Stroke                                 | 78 993<br>(13 RCTs)   | RR = 0.88<br>(0.81–0.96)    | 39 per 1000               | 34 per 1000<br>(32–37)             | Five fewer per 1000<br>(from 7 fewer to 2 fewer)           | ⊕⊕⊕⊕<br>High    |
| HHF                                    | 52 887<br>(10 RCTs)   | RR = 0.93<br>(0.85–1.01)    | 53 per 1000               | 49 per 1000<br>(45–54)             | Four fewer per 1000<br>(from 8 fewer to 1 more)            | ⊕⊕⊕<br>Moderate |
| Coronary revascularization             | 41 099<br>(5 RCTs)    | RR = 0.87<br>(0.74–1.01)    | 126 per 1000              | 110 per 1000<br>(93–127)           | Sixteen fewer per 1000<br>(from 33 fewer to 1 more)        | ⊕⊕⊕<br>Moderate |
| Hospitalization due to unstable angina | 43 604<br>(7 RCTs)    | RR = 0.99<br>(0.86–1.16)    | 27 per 1000               | 27 per 1000<br>(24 to 32)          | Zero fewer per 1000<br>(from 4 fewer to 4 more)            | ⊕⊕⊕<br>Moderate |
| Composite renal outcome                | 41 332<br>(8 RCTs)    | RR = 0.80<br>(0.73–0.88)    | 113 per 1000              | Ninety-one per 1000<br>(83 to 100) | Twenty-three fewer per 1000<br>(from 31 fewer to 14 fewer) | ⊕⊕⊕⊕<br>High    |

CI, confidence interval; CV, cardiovascular; GLP-1 RA, glucagon-like peptide-1 receptor agonist; HHF, hospitalization for heart failure; MACE, major adverse cardiovascular events; N, number; RCT, randomized controlled trial; RR, risk ratio.

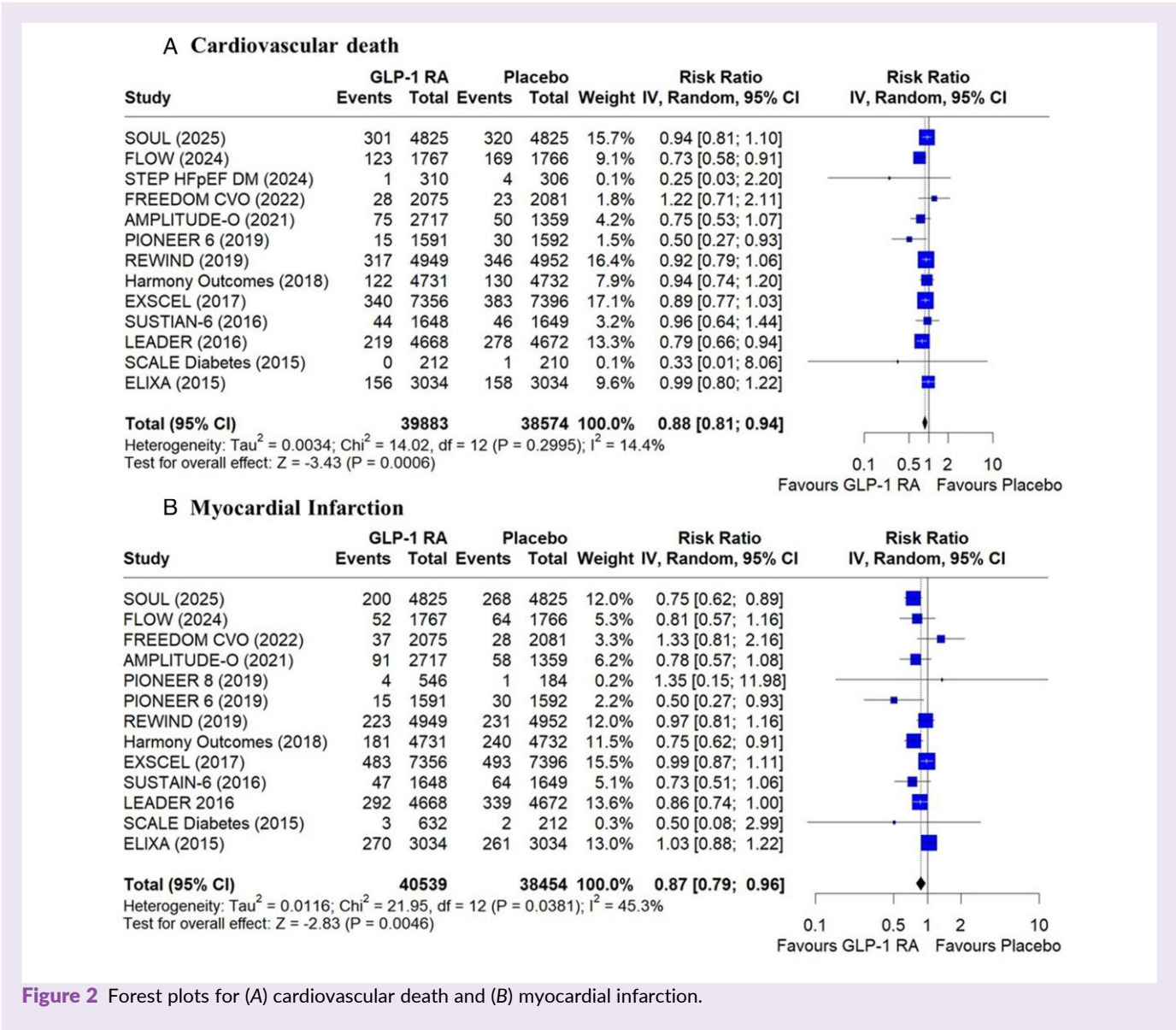
significantly reduced stroke incidence, though MI and CV death were not significantly reduced, possibly due to the primary prevention population.<sup>44</sup> The observed reduction in MACE, stroke, and MI associated with GLP-1RAs can be attributed to several biological mechanisms. Nagashima *et al.* showed GLP-1 and GIP reduced foam cell formation by suppressing CD36 and ACAT1 expression in macrophages.<sup>45</sup> GLP-1RAs also enhance endothelial function by increasing nitric oxide availability, promoting vasodilation, and improving overall vascular tone.<sup>15</sup>

Third, our analysis revealed a trend towards a reduction in coronary revascularization, which became statistically significant in a sensitivity analysis restricted to trials reporting HRs. However, this exploratory finding should be interpreted with caution, given the borderline statistical significance in the primary analysis. This observation is consistent with previous large CVOTs by Marso *et al.* and McGuire *et al.*<sup>15,33</sup> though Patoulas *et al.* did not observe this benefit, possibly due to the inclusion of observational studies.<sup>46</sup> The potential mechanism may involve anti-atherosclerotic effects of GLP-1RAs, including enhanced endothelial function via PKA, PI3 K/Akt, and eNOS signalling

pathways,<sup>47,48</sup> as well as improvements in indirect CV risk factors such as blood pressure, lipids, and inflammation.<sup>49–51</sup>

Fourth, our analysis revealed a modest, non-significant trend towards reduced HHF. It is important to distinguish this small, heterogeneous effect from the robust benefits observed for atherosclerotic outcomes. The phenotype-specific heterogeneity is important: while GLP-1RAs demonstrate promising benefits in HFpEF as evidenced by STEP-HFpEF DM and prior pooled cumulative analyses,<sup>17,52</sup> evidence for HFrEF remains limited and uncertain. Phase 2 HFrEF trials (FIGHT and LIVE) that enrolled patients with or without T2DM failed to demonstrate clinical benefit with liraglutide and raised concerns regarding increased heart rate and serious cardiac adverse events.<sup>53,54</sup> Given the lack of overall benefit and phenotype-specific differences, particularly in HFrEF populations, further research is needed to clarify their safety and efficacy in this subgroup.

Fifth, our analysis showed a significant reduction in composite renal outcomes with GLP-1RA therapy. Renal protection was also observed in SUSTAIN 6 and LEADER among patients with



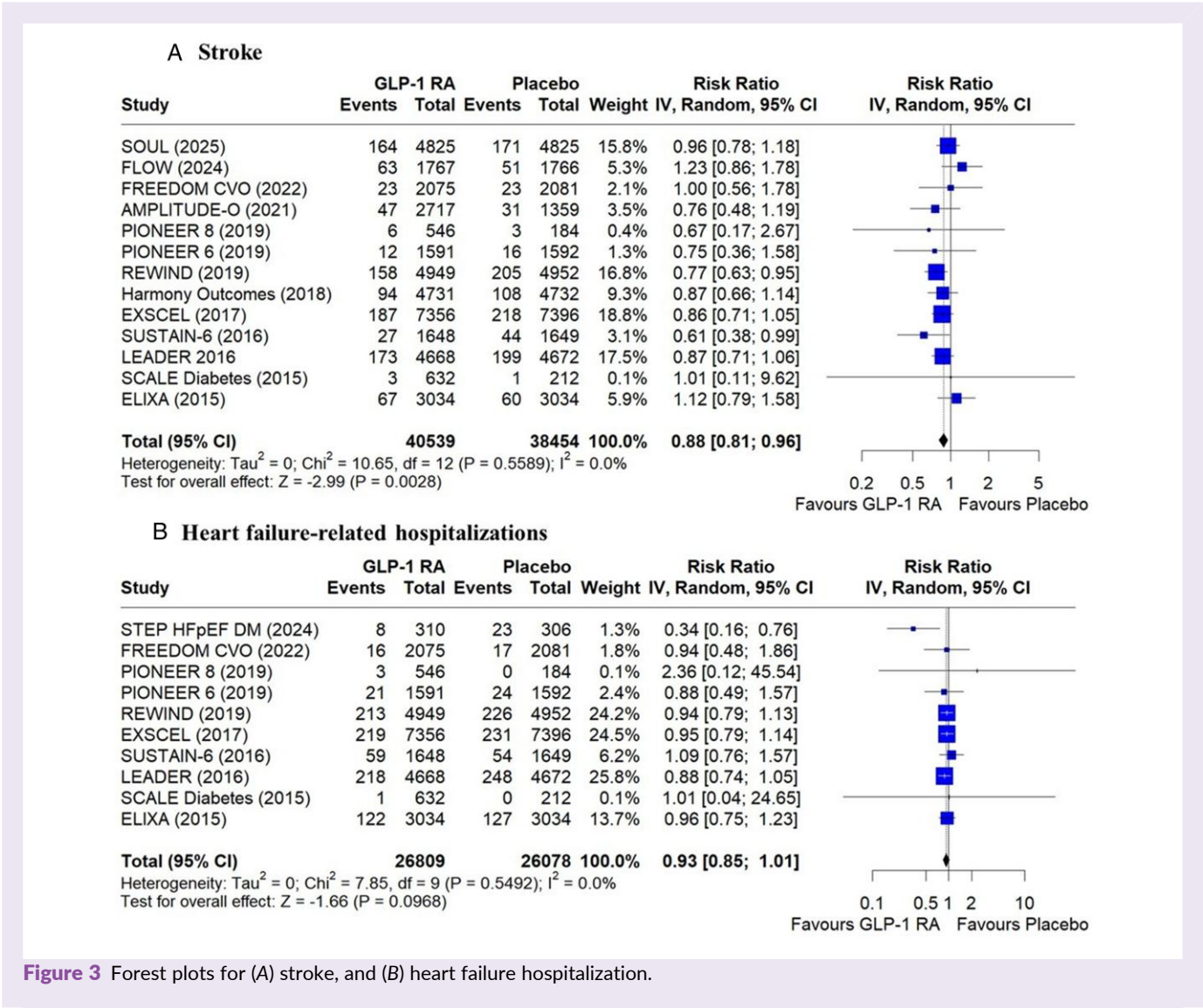
CKD.<sup>32,33</sup> Although the underlying mechanisms of GLP-1RA-mediated nephroprotection are not yet fully understood, several mechanisms have been proposed. These include inhibition of the Na–H exchanger 3 in proximal tubular cells, modulation of afferent arteriolar tone, and enhancement of natriuresis and tubuloglomerular feedback.<sup>55</sup> Whether these mechanisms remain active in advanced kidney disease is uncertain.

While several outcomes demonstrated low statistical heterogeneity (I<sup>2</sup> = 0% for all-cause death, stroke, HFH, and hospitalization due to unstable angina), it is important to recognize that an I<sup>2</sup> of 0% does not definitively exclude heterogeneity; rather, it indicates that heterogeneity cannot be statistically quantified. This apparent statistical homogeneity can mask substantial underlying clinical and methodological diversity that fundamentally limits the causal interpretation of our pooled estimates. Over the timeline of the included trials, standard-of-care background therapies evolved significantly. For example, the use of SGLT2 inhibitors and novel lipid-lowering agents increased markedly in more recent trials, creating a shifting baseline risk that may confound the isolation of the GLP-1RA treatment effect. Additionally, the definition and adjudication of

endpoints, particularly HHF and composite renal outcomes, varied across different trial eras and steering committees, introducing potential misclassification bias. Consequently, these temporal, pharmacological, and methodological shifts mean that the pooled relative risk reductions represent an averaged effect across heterogeneous clinical contexts, rather than a precise, uniform causal estimate applicable to a standardized patient population.

Clinical implications

The findings of this comprehensive meta-analysis bear significant clinical implications for the management of patients with T2DM. The reduction in MACE, all-cause mortality, and cardiovascular mortality reinforces the role of GLP-1RAs not merely as glucose-lowering agents, but as foundational disease-modifying therapies. Current clinical guidelines from major cardiovascular and endocrine societies advocate for the early initiation of GLP-1RAs in patients with T2DM who have established cardiovascular disease or are at high cardiovascular risk, independent of baseline glycated haemoglobin levels [<sup>56,57</sup>]. Our data strongly support this paradigm shift



towards a cardioprotective treatment strategy. Furthermore, the observed benefits in reducing composite renal outcomes highlight the pleiotropic effects of GLP-1RAs, providing a dual mechanism of cardiorenal protection. While the reductions in HHF and coronary revascularizations did not reach statistical significance, the favourable trends suggest that GLP-1RAs may offer comprehensive cardiovascular risk mitigation. Consequently, clinicians should prioritize the integration of GLP-1RAs into the therapeutic regimens for T2DM, ensuring that treatment goals extend beyond glycemic control to encompass broad cardiovascular and renal risk reduction.

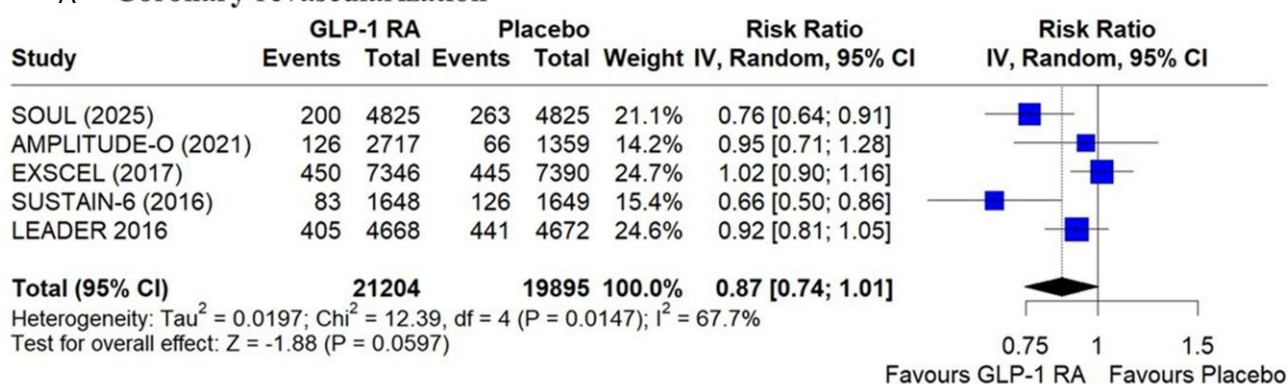
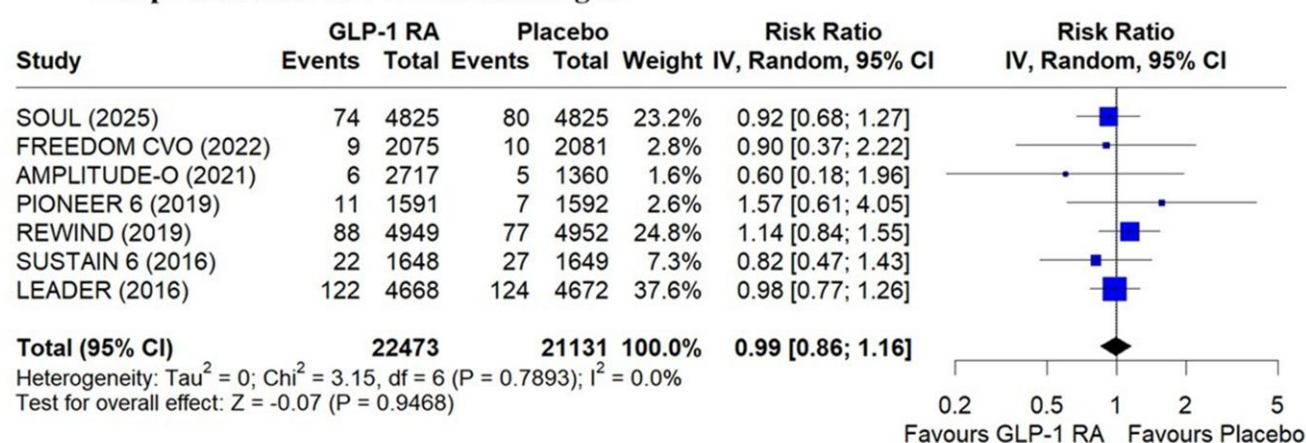
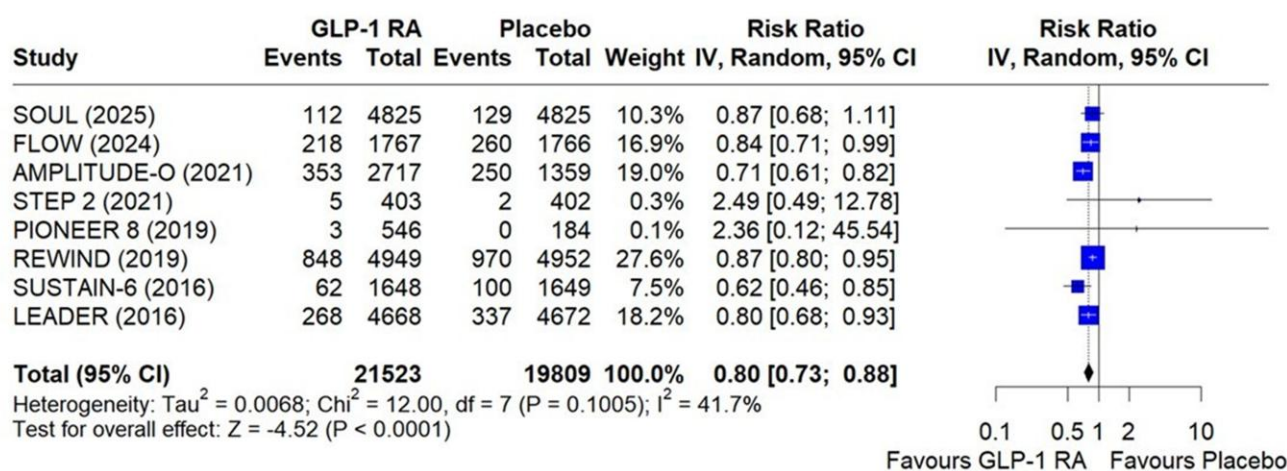
Strengths

This study possesses several notable strengths that enhance the validity and generalizability of its findings. Foremost is the comprehensive nature of the meta-analysis, which pooled data from 20 RCTs encompassing a substantial cohort of 83 004 patients. This large sample size provides a substantial statistical power to detect meaningful differences in both primary and secondary cardiovascular endpoints. Additionally, the inclusion of the most recent CVOTs ensures that the analysis reflects the contemporary therapeutic landscape. The methodological rigour

is further strengthened by the application of the GRADE framework, which confirmed a high certainty of evidence for the primary outcomes of MACE, all-cause death, and cardiovascular death. Furthermore, our analysis extended beyond traditional cardiovascular endpoints to evaluate a broader spectrum of clinically relevant outcomes, including HHF, coronary revascularization, and composite renal endpoints. The calculation of absolute risk reductions alongside relative effect measures provides a more translatable metric for clinical decision-making, allowing practitioners to better weigh the clinical benefits of GLP-1RA therapy in their patient populations. Furthermore, we conducted comprehensive sensitivity analyses, including leave-one-out and HRs-restricted evaluations, which confirmed the stability of the pooled estimates across different follow-up durations. The extensive subgroup analyses provide insights into treatment effects across diverse clinical scenarios, confirming consistent benefits across most patient strata.

Future perspectives

While the cardiovascular benefits of GLP-1RAs in T2DM are now well-established, several avenues for future research

**A Coronary revascularization****B Hospitalizations due to unstable angina****C Composite renal outcome****Figure 4** Forest plots for (A) coronary revascularization, (B) hospitalization due to unstable angina, and (C) composite renal outcome.

remain imperative to optimize their clinical utility. Future prospective trials should be specifically designed to evaluate the efficacy and safety of GLP-1RAs across distinct heart failure phenotypes, particularly HFrEF, where current evidence remains equivocal and potentially confounded by the chronotropic effects of these agents. Additionally, there is a need for studies exploring the comparative effectiveness of different GLP-1RA agents and novel dual or triple incretin co-agonists

(e.g. tirzepatide, retatrutide) to determine if synergistic receptor activation confers superior cardiovascular or renal protection. Further research should also focus on identifying specific patient subgroups defined by genetic, biomarker, or phenotypic profiles who may derive the greatest benefit from GLP-1RA therapy, thereby advancing the principles of precision medicine. Finally, real-world observational studies and long-term follow-up of existing trial cohorts are necessary to assess the durability

of these cardioprotective effects, long-term safety profiles, and the cost-effectiveness of widespread GLP-1RA implementation in diverse, unselected clinical populations.<sup>58,59</sup>

## Limitations

This meta-analysis, though comprehensive, has several limitations. First, considerable heterogeneity existed among included trials regarding design, follow-up duration, participant characteristics, and types/doses of GLP-1RAs, which may impact the generalizability of findings. Second, variation in endpoint definitions—especially for MACE, HHF, and renal outcomes may have introduced misclassification and contributed to heterogeneity. Third, the CI widths varied considerably across trials, reflecting differences in sample size and event rates. Large CV outcome trials contributed most of the statistical weight and yielded more precise estimates, whereas smaller or underpowered studies had wide CIs and limited influence on the pooled results, warranting cautious interpretation. Although a subgroup analysis based on trial size was performed to explore this variability, no significant interaction was observed for the primary outcomes. Fourth, several deviations from the registered PROSPERO protocol should be acknowledged. While the protocol specified the use of both RevMan and R software for statistical analyses, only R was ultimately employed, as it provided more comprehensive functionality for the meta-analytic approaches required for this review. Additionally, the screening process was conducted using Rayyan rather than the initially planned Google Sheets approach, as Rayyan offered superior collaboration features and systematic review management capabilities. These methodological adaptations were made to enhance the rigour and efficiency of the review process but were not pre-specified in the protocol. Lastly, the lack of individual patient data (IPD) limited detailed subgroup analyses (e.g. by age, sex, cardiovascular risk, renal function, background medical therapy or HF phenotype) and adjustment for confounders influencing treatment response.

## Conclusion

Our study highlights the significant cardiometabolic benefits of GLP-1RAs in individuals with T2DM. GLP-1RAs were associated with a significant reduction in MACE, all-cause and CV mortality, MI, stroke, and composite renal outcomes. While a trend towards a reduction in heart failure hospitalizations and coronary revascularization was observed, these findings did not reach statistical significance. Various biological mechanisms, including anti-inflammatory, anti-atherosclerotic, and metabolic effects, support these benefits. Overall, GLP-1RAs demonstrate a robust profile for cardiovascular and renal protection and represent an important therapeutic option in the management of patients with T2DM.

## Supplementary material

Supplementary material is available at *European Heart Journal—Quality of Care and Clinical Outcomes* online.

## Author contributions

Mushood Ahmed [Data curation, Formal analysis, Writing—original draft, Writing—review & editing (lead)], Tallal Mushtaq Hashmi [Conceptualization, Data curation, Writing—original draft, Writing—review & editing (supporting)], Maryam Shahzad [Conceptualization, Data curation, Writing—review & editing

(equal)], Rohma Zia [Conceptualization, Data curation, Writing—review & editing (equal)], Hadiyah Ashraf [Conceptualization, Data curation, Writing—original draft (supporting)], Raheel Ahmed [Validation, Writing—original draft, Writing—review & editing (supporting)], Muhammad Rashid (Conceptualization, Data curation, Methodology, Writing—original draft [lead], Formal analysis, Writing—review & editing [equal]), Stephen J. Greene [Writing—original draft, Writing—review & editing (equal)], Robert J. Mentz [Writing—original draft, Writing—review & editing (equal)], Marat Fudim [Conceptualization, Writing—original draft (equal)], Gregg C. Fonarow [Writing—original draft, Writing—review & editing (equal)], and Jamal S Rana [Writing—original draft, Writing—review & editing (lead)]

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## Data availability

All data generated or analysed during this study are included in this article. Further inquiries can be directed to the corresponding author.

## Ethical approval

No ethical approval was required for the study.

## References

1. Lascar N, Brown J, Pattison H, Barnett AH, Bailey CJ, Bellary S. Type 2 diabetes in adolescents and young adults. *Lancet Diabetes Endocrinol* 2018;**6**:69–80. [https://doi.org/10.1016/S2213-8587\(17\)30186-9](https://doi.org/10.1016/S2213-8587(17)30186-9)
2. Mlynarska E, Czarnik W, Dzieża N, Jędraszak W, Majchrowicz G, Prusinowski F, et al. Type 2 diabetes mellitus: new pathogenetic mechanisms, treatment and the most important complications. *Int J Mol Sci* 2025;**26**:1094. <https://doi.org/10.3390/ijms26031094>
3. Ahmed M, Zulfikar E, Shafiq A, Shahzad M, Hashmi TM, Ahmed R, et al. Type 2 diabetes Mellitus-related mortality in the United States, 1999 to 2023. *JACC Adv* 2025;**4**:101882. <https://doi.org/10.1016/j.jaccadv.2025.101882>
4. Menke A, Casagrande S, Geiss L, Cowie CC. Prevalence of and trends in diabetes among adults in the United States, 1988–2012. *JAMA* 2015;**314**:1021–1029. <https://doi.org/10.1001/jama.2015.10029>
5. Chong K, Chang JK-J, Chuang L-M. Recent advances in the treatment of type 2 diabetes mellitus using new drug therapies. *Kaohsiung J Med Sci* 2024;**40**:212–220. <https://doi.org/10.1002/kjm2.12800>
6. Ahmed M, Nofal A, Shafiq A, Javaid H, Ahsan A, Nadeem ZA, et al. Rising mortality rates linked to type-2 diabetes and obesity in the United States: an observational analysis from 1999 to 2022. *J Diabetes Investig* 2025;**16**:492–500. <https://doi.org/10.1111/jdi.14386>
7. Garber AJ, Abrahamson MJ, Barzilay JI, Blonde L, Bloomgarden ZT, Bush MA, et al. American Association of Clinical Endocrinologists' comprehensive diabetes management algorithm 2013 consensus statement—executive summary. *Endocr Pract* 2013;**19**:536–557. <https://doi.org/10.4158/EP13176.CS>
8. Marx N, Husain M, Lehrke M, Verma S, Sattar N. GLP-1 Receptor agonists for the reduction of atherosclerotic cardiovascular risk in patients with type 2 diabetes. *Circulation* 2022;**146**:1882–1894. <https://doi.org/10.1161/CIRCULATIONAHA.122.059595>
9. Brown E, Rajeev SP, Cuthbertson DJ, Wilding JPH. A review of the mechanism of action, metabolic profile and haemodynamic effects of sodium-glucose co-transporter-2 inhibitors. *Diabetes Obes Metab* 2019;**21**:9–18. <https://doi.org/10.1111/dom.13650>
10. Kristensen SL, Rørth R, Jhund PS, Docherty KF, Sattar N, Preiss D, et al. Cardiovascular, mortality, and kidney outcomes with GLP-1 receptor agonists in patients with type 2 diabetes: a systematic review and meta-analysis of cardiovascular outcome trials. *Lancet Diabetes Endocrinol* 2019;**7**:776–785. [https://doi.org/10.1016/S2213-8587\(19\)30249-9](https://doi.org/10.1016/S2213-8587(19)30249-9)
11. Zelniker TA, Wiviott SD, Raz I, Im K, Goodrich EL, Bonaca MP, et al. SGLT2 inhibitors for primary and secondary prevention of cardiovascular and renal outcomes in type 2 diabetes: a systematic review and meta-analysis of cardiovascular outcome trials. *Lancet* 2019;**393**:31–39. [https://doi.org/10.1016/S0140-6736\(18\)32590-X](https://doi.org/10.1016/S0140-6736(18)32590-X)
12. Zelniker TA, Wiviott SD, Raz I, Im K, Goodrich EL, Furtado RHM, et al. Comparison of the effects of glucagon-like peptide receptor agonists and sodium-glucose cotransporter 2 inhibitors for prevention of major adverse cardiovascular and renal outcomes in type 2 diabetes mellitus. *Circulation* 2019;**139**:2022–2031. <https://doi.org/10.1161/CIRCULATIONAHA.118.038868>
13. Nauck MA, Meier JJ. Management of endocrine disease: are all GLP-1 agonists equal in the treatment of type 2 diabetes? *Eur J Endocrinol* 2019;**181**:R211–R234. <https://doi.org/10.1530/EJE-19-0566>
14. Shin J-I. Second-line glucose-lowering therapy in type 2 diabetes mellitus. *Curr Diab Rep* 2019;**19**:54. <https://doi.org/10.1007/s11892-019-1171-0>
15. McGuire DK, Marx N, Mulvagh SL, Deanfield JE, Inzucchi SE, Pop-Busui R, et al. Oral semaglutide and cardiovascular outcomes in high-risk type 2 diabetes. *N Engl J Med* 2025;**392**:2001–2012. <https://doi.org/10.1056/NEJMoa2501006>
16. Perkovic V, Tuttle KR, Rossing P, Mahaffey KW, Mann JFE, Bakris G, et al. Effects of semaglutide on chronic kidney disease in patients with type 2 diabetes. *N Engl J Med* 2024;**391**:109–121. <https://doi.org/10.1056/NEJMoa2403347>
17. Kosiborod MN, Petrie MC, Borlaug BA, Butler J, Davies MJ, Hovingh GK, et al. Semaglutide in patients with obesity-related heart failure and type 2 diabetes. *N Engl J Med* 2024;**390**:1394–1407. <https://doi.org/10.1056/NEJMoa2313917>
18. Bonaca MP, Catarig A-M, Houlind K, Ludvik B, Nordanstig J, Ramesh CK, et al. Semaglutide and walking capacity in people with symptomatic peripheral artery disease and type 2 diabetes (STRIDE): a phase 3b, double-blind, randomised, placebo-controlled trial. *Lancet* 2025;**405**:1580–1593. [https://doi.org/10.1016/S0140-6736\(25\)00509-4](https://doi.org/10.1016/S0140-6736(25)00509-4)
19. Kelly M, Lewis J, Rao H, Carter J, Portillo I, Beuttler R. Effects of GLP-1 receptor agonists on cardiovascular outcomes in patients with type 2 diabetes and chronic kidney disease: a systematic review and meta-analysis. *Pharmacotherapy* 2022;**42**:921–928. <https://doi.org/10.1002/phar.2737>
20. Lee MMY, Sattar N, Pop-Busui R, Deanfield J, Emerson SS, Inzucchi SE, et al. Cardiovascular and kidney outcomes and mortality with long-acting injectable and oral glucagon-like peptide 1 receptor agonists in individuals with type 2 diabetes: a systematic review and meta-analysis of randomized trials. *Diabetes Care* 2025;**48**:846–859. <https://doi.org/10.2337/dc25-0241>
21. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. <https://doi.org/10.1136/bmj.n71>
22. Sterne JAC, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. Rob 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**366**:l4898. <https://doi.org/10.1136/bmj.l4898>
23. Ruff CT, Baron M, Im K, O'Donoghue ML, Fiedorek FT, Sabatine MS. Subcutaneous infusion of exenatide and cardiovascular outcomes in type 2 diabetes: a non-inferiority randomized controlled trial. *Nat Med* 2022;**28**:89–95. <https://doi.org/10.1038/s41591-021-01584-3>
24. Gerstein HC, Sattar N, Rosenstock J, Ramasundarahettige C, Pratley R, Lopes RD, et al. Cardiovascular and renal outcomes with efpeglenatide in type 2 diabetes. *N Engl J Med* 2021;**385**:896–907. <https://doi.org/10.1056/NEJMoa2108269>
25. Davies M, Færch L, Jeppesen OK, Pakseresht A, Pedersen SD, Perreault L, et al. Semaglutide 2.4 mg once a week in adults with overweight or obesity, and type 2 diabetes (STEP 2): a randomised, double-blind, double-dummy, placebo-controlled, phase 3 trial. *Lancet* 2021;**397**:971–984. [https://doi.org/10.1016/S0140-6736\(21\)00213-0](https://doi.org/10.1016/S0140-6736(21)00213-0)
26. Zinman B, Aroda VR, Buse JB, Cariou B, Harris SB, Hoff ST, et al. Efficacy, safety, and tolerability of oral semaglutide versus placebo added to insulin with or without metformin in patients with type 2 diabetes: the PIONEER 8 trial. *Diabetes Care* 2019;**42**:2262–2271. <https://doi.org/10.2337/dc19-0898>
27. Husain M, Birkenfeld AL, Donsmark M, Dungan K, Eliaschewitz FG, Franco DR, et al. Oral semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *N Engl J Med* 2019;**381**:841–851. <https://doi.org/10.1056/NEJMoa1901118>
28. Gerstein HC, Colhoun HM, Dagenais GR, Diaz R, Lakshmanan M, Pais P, et al. Dulaglutide and cardiovascular outcomes in type 2 diabetes (REWIND): a double-blind, randomised placebo-controlled trial. *Lancet* 2019;**394**:121–130. [https://doi.org/10.1016/S0140-6736\(19\)31149-3](https://doi.org/10.1016/S0140-6736(19)31149-3)
29. Hernandez AF, Green JB, Janmohamed S, D'Agostino RB, Granger CB, Jones NP, et al. Albiglutide and cardiovascular outcomes in patients with type 2 diabetes and cardiovascular disease (harmony outcomes): a double-blind, randomised placebo-controlled trial. *Lancet* 2018;**392**:1519–1529. [https://doi.org/10.1016/S0140-6736\(18\)32261-X](https://doi.org/10.1016/S0140-6736(18)32261-X)
30. Holman RR, Bethel MA, Mentz RJ, Thompson VP, Lokhnygina Y, Buse JB, et al. Effects of once-weekly exenatide on cardiovascular outcomes in type 2 diabetes. *N Engl J Med* 2017;**377**:1228–1239. <https://doi.org/10.1056/NEJMoa1612917>
31. Nauck MA, Stewart MW, Perkins C, Jones-Leone A, Yang F, Perry C, et al. Efficacy and safety of once-weekly GLP-1 receptor agonist albiglutide (HARMONY 2): 52 week primary endpoint results from a randomised, placebo-controlled trial in patients with type 2 diabetes mellitus inadequately controlled with diet and exercise. *Diabetologia* 2016;**59**:266–274. <https://doi.org/10.1007/s00125-015-3795-1>
32. Marso SP, Bain SC, Consoi A, Eliaschewitz FG, Jódar E, Leiter LA, et al. Semaglutide and cardiovascular outcomes in patients with type 2 diabetes. *N Engl J Med* 2016;**375**:1834–1844. <https://doi.org/10.1056/NEJMoa1607141>
33. Marso SP, Daniels GH, Brown-Frandsen K, Kristensen P, Mann JFE, Nauck MA, et al. Liraglutide and cardiovascular outcomes in type 2 diabetes. *N Engl J Med* 2016;**375**:311–322. <https://doi.org/10.1056/NEJMoa1603827>
34. Davies MJ, Bergenstal R, Bode B, Kushner RF, Lewin A, Skjøth TV, et al. Efficacy of liraglutide for weight loss among patients with type 2 diabetes: the SCALE diabetes randomized clinical trial. *JAMA* 2015;**314**:687–699. <https://doi.org/10.1001/jama.2015.9676>
35. Pfeffer MA, Claggett B, Diaz R, Dickstein K, Gerstein HC, Køber LV, et al. Lixisenatide in patients with type 2 diabetes and acute coronary syndrome. *N Engl J Med* 2015;**373**:2247–2257. <https://doi.org/10.1056/NEJMoa1509225>
36. Reusch J, Stewart MW, Perkins CM, Cirkel DT, Ye J, Perry CR, et al. Efficacy and safety of once-weekly glucagon-like peptide 1 receptor agonist albiglutide (HARMONY 1 trial): 52-week primary endpoint results from a randomized, double-blind, placebo-controlled trial in patients with type 2 diabetes mellitus not controlled on pioglitazone, with or without metformin. *Diabetes Obes Metab* 2014;**16**:1257–1264. <https://doi.org/10.1111/dom.12382>
37. Home PD, Shamanna P, Stewart M, Yang F, Miller M, Perry C, et al. Efficacy and tolerability of albiglutide versus placebo or pioglitazone over 1 year in people with type 2 diabetes currently taking metformin and glimepiride: HARMONY

5. *Diabetes Obes Metab* 2015;17:179–187. <https://doi.org/10.1111/dom.12414>
38. Åhrén B, Johnson SL, Stewart M, Cirkel DT, Yang F, Perry C, et al. HARMONY 3: 104-week randomized, double-blind, placebo- and active-controlled trial assessing the efficacy and safety of albiglutide compared with placebo, sitagliptin, and glimepiride in patients with type 2 diabetes taking metformin. *Diabetes Care* 2014;37:2141–2148. <https://doi.org/10.2337/dc14-0024>
39. Wilbon SS, Kolonin MG. GLP1 receptor agonists-effects beyond obesity and diabetes. *Cells* 2023;13:65. <https://doi.org/10.3390/cells13010065>
40. Lecis D, Prandi FR, Barone L, Belli M, Sergi D, Longo S, et al. Beyond the cardiovascular effects of glucagon-like peptide-1 receptor agonists: body slimming and plaque stabilization. Are new statins born? *Biomolecules* 2023;13:1695. <https://doi.org/10.3390/biom13121695>
41. Sun F, Wu S, Wang J, Guo S, Chai S, Yang Z, et al. Effect of glucagon-like peptide-1 receptor agonists on lipid profiles among type 2 diabetes: a systematic review and network meta-analysis. *Clin Ther* 2015;37:22–5–241.e8. <https://doi.org/10.1016/j.clinthera.2014.11.008>
42. Rivera FB, Lumbang GNO, Gaid DRM, Cruz LLA, Magalong JV, Bantayan NRB, et al. Glucagon-like peptide-1 receptor agonists modestly reduced blood pressure among patients with and without diabetes mellitus: a meta-analysis and meta-regression. *Diabetes Obes Metab* 2024;26:2209–2228. <https://doi.org/10.1111/dom.15529>
43. Alicic RZ, Cox EJ, Neumiller JJ, Tuttle KR. Incretin drugs in diabetic kidney disease: biological mechanisms and clinical evidence. *Nat Rev Nephrol* 2021;17:227–244. <https://doi.org/10.1038/s41581-020-00367-2>
44. Rakipovski G, Rolin B, Nøhr J, Klewe I, Frederiksen KS, Augustin R, et al. The GLP-1 analogs liraglutide and semaglutide reduce atherosclerosis in ApoE<sup>-/-</sup> and LDLr<sup>-/-</sup> mice by a mechanism that includes inflammatory pathways. *JACC Basic Transl Sci* 2018;3:844–857. <https://doi.org/10.1016/j.jacbs.2018.09.004>
45. Nagashima M, Watanabe T, Terasaki M, Tomoyasu M, Nohtomi K, Kim-Kaneyama J, et al. Native incretins prevent the development of atherosclerotic lesions in apolipoprotein E knockout mice. *Diabetologia* 2011;54:2649–2659. <https://doi.org/10.1007/s00125-011-2241-2>
46. Patoulas D, Papadopoulos C, Kassimis G, Doumas M. Glucagon-like peptide-1 receptor agonists do not decrease the risk of coronary revascularization in patients with type 2 diabetes. *Coron Artery Dis* 2022;33:427–428. <https://doi.org/10.1097/MCA.0000000000001111>
47. Erdogdu O, Nathanson D, Sjöholm A, Nyström T, Zhang Q. Exendin-4 stimulates proliferation of human coronary artery endothelial cells through eNOS-, PKA- and PI3K/Akt-dependent pathways and requires GLP-1 receptor. *Mol Cell Endocrinol* 2010;325:26–35. <https://doi.org/10.1016/j.mce.2010.04.022>
48. Kapadia P, Bikkina P, Landicho MA, Parekh S, Haas MJ, Mooradian AD. Effect of anti-hyperglycemic drugs on endoplasmic reticulum (ER) stress in human coronary artery endothelial cells. *Eur J Pharmacol* 2021;907:174249. <https://doi.org/10.1016/j.ejphar.2021.174249>
49. Trujillo JM, Nuffer W, Smith BA. GLP-1 receptor agonists: an updated review of head-to-head clinical studies. *Ther Adv Endocrinol Metab* 2021;12:2042018821997320. <https://doi.org/10.1177/2042018821997320>
50. Sardu C, Paolisso P, Sacra C, Santamaria M, de Lucia C, Ruocco A, et al. Cardiac resynchronization therapy with a defibrillator (CRTd) in failing heart patients with type 2 diabetes mellitus and treated by glucagon-like peptide 1 receptor agonists (GLP-1 RA) therapy vs. conventional hypoglycemic drugs: arrhythmic burden, hospitalizations for heart failure, and CRTd responders rate. *Cardiovasc Diabetol* 2018;17:137. <https://doi.org/10.1186/s12933-018-0778-9>
51. Sternkopf M, Nagy M, Baaten CCFMJ, Kuipers MJE, Tullemans BME, Wirth J, et al. Native, intact glucagon-like peptide 1 is a natural suppressor of thrombus growth under physiological flow conditions. *Arterioscler Thromb Vasc Biol* 2020;40:e65–e77. <https://doi.org/10.1161/ATVBAHA.119.313645>
52. Ahmed M, Burhan M, Shafiq A, Hashmi TM, Ahmed R, Fudim M, et al. GLP-1 and dual GLP-1/GIP receptor agonists in heart failure with mildly reduced or preserved ejection fraction: a systematic review and meta-analysis. *Clin Cardiol* 2025;48:e70234. <https://doi.org/10.1002/clc.70234>
53. Margulies KB, Hernandez AF, Redfield MM, Givertz MM, Oliveira GH, Cole R, et al. Effects of liraglutide on clinical stability among patients with advanced heart failure and reduced ejection fraction: a randomized clinical trial. *JAMA* 2016;316:500–508. <https://doi.org/10.1001/jama.2016.10260>
54. Jorsal A, Kistorp C, Holmager P, Tougaard RS, Nielsen R, Hänselmann A, et al. Effect of liraglutide, a glucagon-like peptide-1 analogue, on left ventricular function in stable chronic heart failure patients with and without diabetes (LIVE)—a multicentre, double-blind, randomised, placebo-controlled trial. *Eur J Heart Fail* 2017;19:69–77. <https://doi.org/10.1002/ehf.657>
55. Hviid AVR, Sørensen CM. Glucagon-like peptide-1 receptors in the kidney: impact on renal autoregulation. *Am J Physiol Renal Physiol* 2020;318:F443–F454. <https://doi.org/10.1152/ajprenal.00280.2019>
56. American Diabetes Association Professional Practice Committee. 9. Pharmacologic approaches to glycemic treatment: standards of care in diabetes—2025. *Diabetes Care* 2024;48:S181–S206. <https://doi.org/10.2337/dc25-S009>
57. Marx N, Federici M, Schütt K, Müller-Wieland D, Ajjan RA, Antunes MJ, et al. 2023 ESC guidelines for the management of cardiovascular disease in patients with diabetes. *Eur Heart J* 2023;44:4043–4140. <https://doi.org/10.1093/eurheartj/ehad192>
58. Kishimori T, Kato T, Wada A, Tani A, Yamaji R, Koike J, et al. Cardiovascular outcomes and safety of semaglutide in non-overweight populations with type 2 diabetes: a comparison with dipeptidyl peptidase 4 inhibitors. *Eur Heart J Qual Care Clin Outcomes* 2025;11:1319–1328. <https://doi.org/10.1093/ehjqcco/qcaf065>
59. Zomer E, Zhou J, Nelson AJ, Sumithran P, Nanayakkara S, Ball J, et al. The cost-effectiveness of semaglutide in reducing cardiovascular risk among people with overweight and obesity and existing cardiovascular disease, but without diabetes. *Eur Heart J Qual Care Clin Outcomes* 2025;11:857–867. <https://doi.org/10.1093/ehjqcco/qcae063>